

J. AMIEL M. FLORES

jamflores@ucla.edu
amielflores.wixsite.com/mysite

EDUCATION

- | | | |
|------------|---|----------------------|
| PhD | Ecology and Evolutionary Biology C. Phil.
University of California, Los Angeles
Advisor: Dr. Paul Barber
GPA: 4.0 | Sept. 2021 – Present |
| BS | Evo. Eco. & Biodiv. with Honors
University of California, Davis
Mentors: Dr. Rachael Bay, Dr. Anne Todgham
GPA: 3.9 | Sept. 2021 – Present |

RESEARCH EXPERIENCE

- | | |
|--|-----------------------|
| Barber Lab , University of California, Los Angeles
Advisor: Dr. Paul Barber
Dissertation: <i>Epibiont Community Compositions on Floating Turbinaria Rafts, Novel Ecosystems in an Increasingly Stormy Anthropocene</i> <ul style="list-style-type: none">• Utilizing metabarcoding and metagenomics to describe the epibiont communities on burgeoning <i>Turbinaria</i> rafts in the South Pacific and understanding their potential as vectors of dispersal for toxic dinoflagellates, coral pathogens, microorganisms, and other epibionts.• Presented findings at conferences.• Collaborated with graduate students from UC, Santa Barbara and University of French Polynesia to communicate research which is also used for Tahitian-English language learning.• Mentored 4 students that have graduated; one of my mentees is now in a PhD program.• One of my current mentees is currently pursuing an independent research project on Southern California seagrass microbiomes. | Sept. 2021 – Present |
| The Diversity Project , University of California, Los Angeles
Advisors: Dr. Paul Barber and Dr. Peggy Fong <ul style="list-style-type: none">• Collaborated with other researchers explore the effects of simulated storm disturbance of differing magnitude on the short term ‘transient’ dynamics of the herbivore fish community.• Conducted field experiments via SCUBA and snorkeling, processed fish behavior videos collected from the field, and analyzed data for presentation.• Earned American Academy of Underwater Sciences (AAUS) Scientific Diver Certifications.• Achieved Master SCUBA Diver Certifications National Association of Underwater Instructors (NAUI).• Presented findings at conferences, symposiums, and as a guest speaker for several UCLA courses. | June 2021 – Aug. 2021 |
| Bay Lab , University of California, Davis
Advisor: Dr. Rachael Bay <ul style="list-style-type: none">• Reviewed and analyzed peer reviewed scientific articles relating to coral biology.• Gathered and analyzed data from available published data for a meta-analysis project identifying genomic diversity in <i>Acropora</i> and <i>Porites</i> species.• Worked in collaboration with other lab members to identify different factors that may contribute to coral genomic diversity.• Gathered environmental data that may potentially be shaping genetic diversity in species of <i>Acropora</i> and <i>Porites</i>.• Presented findings at conferences and symposiums. | Nov. 2019 – June 2021 |

Todgham Lab, University of California, Davis
Advisor: Dr. Anne Todgham

Oct. 2018– June 2021

- Conducted environmental change experiments on western mosquitofish (*Gambusia affinis*).
- Helped in the initial exploration and organization of a mosquitofish project for a graduate student in the lab.
- Helped investigate the effects of temperature and oxygen levels on the early life stages of Chinook salmon (*Oncorhynchus tshawytscha*).
- Created a temperature ramping apparatus for Critical Thermal Maxima Tests.
- Designed, conducted, and presented results from my independent research experiment at symposiums.

TEACHING AND MENTORSHIP EXPERIENCE

Mentor – GradMAP

Aug. 2024 – Present

Assisted mentees in the Philippines through the graduate school application process by providing mentorship and guidance over six months through GradMAP.

Undergraduate Lab Meeting Co-Facilitator – UCLA Barber Lab

Jan. 2024 – Present

Facilitate a space for all undergraduate research assistants in the Barber Lab to check in with the PI and connect with other undergraduates in the lab. Organize discussions pertaining to current and summer research opportunities, upcoming research conferences, graduate school application process.

Graduate Student Mentor – UCLA Barber Lab

Nov. 2023 – Present

Mentoring six undergraduate students of underrepresented backgrounds in marine science in different stages of their undergraduate career at UCLA. The mentor-mentee relationship includes training mentees in the workflow of genetic and genomic laboratory techniques; individual meetings to discuss and reflect on their progress as well as provide support for their aims; team meetings that provide a space for teambuilding and opportunity to discuss ideas for independent research projects. Organized meetings with undergraduates that are interested in gaining research experience, graduate students looking to mentor undergraduates, and the PI.

Graduate Teaching Assistant – Communicating Science about Health Disparities and the Environment (EE BIOL c179 at UCLA)

April. 2023 – June 2023

Guided and provided feedback to groups on their infographics and presentations. Graded and provided feedback on assignments and course projects to students. This course is part of a series that includes EE BIOL 187 at UCLA.

Graduate Teaching Assistant – Researching Health & Pollution Exposure (EE BIOL 187 at UCLA)

Jan. 2023 – March 2023

Facilitated multiple groups through their environmental justice projects addressing lead exposure. Transported and accompanied students to the community where they based their research. Graded and provided feedback on assignments and course projects to students. This course is part of a course series that includes EE BIOL c179 at UCLA)

Graduate Teaching Assistant – Intro to Marine Science (EE BIOL 109 at UCLA)

Sept. 2022 – Dec. 2022

Led students through discussion of assigned articles in order to connect key concepts that were covered in class to literature. Graded and provided feedback on assignments and course projects to students.

Near Peer Mentor – UCLA The Diversity Project

June 2022 – Aug. 2022

Aided trainees through scientific diving training. Mentored trainees through the scientific process and provided connections and resources in marine sciences.

Tutor – UC Davis Academic Assistance and Tutoring Center

Sept. 2020– June 2021

Courses Tutored

- Structure & Function of Biomolecules (BIS 102)
- Organic Chemistry: Brief Course (CHE 8 Series)

Co-Instructor – The BUSP Mindset: First Year Aggie Connection (BIS 98 at UCD)

Sept. 2020 – Dec. 2020

Exposed first year BUSP students through the resources available at UC Davis. Led students through effective organization and study methods to succeed during their undergraduate.

Head Tutor – UC Davis Leaders of Achievement in Biological Sciences

Sept. 2019 – June 2021

Lead problem solving workshops for second year BUSP UC Davis students enrolled in Introductory Biology (BIS 2 series). Organize and attended study sessions for BUSP students. Evaluated and mentored tutors through their problem-solving workshops.

Courses Tutored:

- Introduction to Biology (BIS 2 Series)
 - Essentials of Life on Earth (BIS 2A)
 - Principles of Ecology & Evolution (BIS 2B)
 - Biodiversity & the Tree of Life (BIS 2C)

Peer Mentor – UC Davis Biology Undergraduate Scholars Program

Sept. 2018 – June 2021

Mentor and aided UC Davis BUSP underclassmen on transitioning into university life. Guided students through the introductory chemistry (CHE 2) and calculus (MAT 17) series.

Courses Tutored:

- General Chemistry (CHE 2 Series)
- Calculus for Biology & Medicine (MAT 17 Series)

HONORS AND AWARDS

CDLS Early Career Fellowship (\$2000) UCLA Center for Diverse Leadership in Science	2026
Departmental Research Grant (\$1000) UCLA Graduate Admissions and Support Committee	2025
French American Doctoral Exchange (FADEx) Program Office for Science and Technology at the Embassy of France	2025
CDLS Climate Resilience Fellowship (\$4000) UCLA Center for Diverse Leadership in Science	2025
CDLS Early Career Fellowship (\$4000) UCLA Center for Diverse Leadership in Science	2025
CDLS Climate Resilience Fellowship (\$2000) UCLA Center for Diverse Leadership in Science	2024
CDLS Early Career Fellowship (\$2000) UCLA Center for Diverse Leadership in Science	2024

Southeast Asia Foreign Language and Area Studies Fellowship (\$8,000) UCLA Center for Southeast Asian Studies	2023
NSF Graduate Research Fellowship Program (\$49,000 per year over 3 years) National Science Foundation	2023
Pauley Fellowship (\$30,000) University of California, Los Angeles	2021
Competitive Edge Fellowship (\$6,000) University of California, Los Angeles	2021
Departmental Citation in Evo., Eco., and Biodiv. UC Davis Department of Evolution and Ecology	2021
Distinguished Scholar Award UC Davis College of Biological Sciences	2021
Barry Goldwater Scholarship Nomination from UC Davis	2020
BUSP Honors Summer Research Fellowship (\$3,240) UC Davis Biology Undergraduate Scholars Program (BUSP)	2019
BUSP Most Outstanding 2nd Year UC Davis Biology Undergraduate Scholars Program	2019
BUSP Most Outstanding 2nd Year UC Davis Biology Undergraduate Scholars Program	2021
UC Davis Dean's Honors List	2017 – 2021
Joannie Mukai Scholarship (\$500)	2017

PUBLICATIONS

Fong C.R., **Flores J.A.**, Hayes H., Williams L., Holbrook S.J., Schmitt R.J., Barber P.H., Fong P. Physical Disturbance to Foundational Macroalgal Communities on Coral Reefs Causes Rapid Changes in Use by Herbivorous Fish Communities. In prep.

Reynolds A.H., Ho E., Steinitz R., Arroyo M., Baldino M., Boslough K., Carlson R., Chavez S.H., Esparza M., Fearon J.E., **Flores J.A.**, Kouba P.V., Sokolov N.A., Payne A. The Queer & Trans Field Safety Assessment: A tool for protecting minoritized field scientists. Pre-print.

PRESENTATIONS, INVITED LECTURES, AND PROFESSIONAL DEVELOPMENT CONFERENCES

Flores, A. and Barber, P.H. Epibiont Community Compositions on Floating *Turbinaria* Rafts, Novel Ecosystems in an Increasingly Stormy Anthropocene. United Nations One Ocean Science Congress 2025, Nice, France, June 3-6, 2025. Oral Presentation. <https://doi.org/10.5194/oos2025-803>, 2025.

Flores, J.A. and Barber, P.H. Epibiont Community Compositions on Floating *Turbinaria* Rafts, Novel Ecosystems in an Increasingly Stormy Anthropocene. Western Society of Naturalists Conference, November 7-11, 2022. Poster Presentation.

Flores, J.A., Hayes, H., Williams, L., Fong, C.R., Holbrook, S.J., Schmitt, R.J., Barber, P.H., Fong, P. Physical Disturbance to Foundational Macroalgal Communities on Coral Reefs Causes Rapid Changes in Use by

Herbivorous Fish Communities. Western Society of Naturalists Conference, November 10-13, 2022. Poster Presentation.

Flores, J.A., Hayes, H., Williams, L., Fong, C.R., Holbrook, S.J., Schmitt, R.J., Barber, P.H., Fong, P. Physical Disturbance to Foundational Macroalgal Communities on Coral Reefs Causes Rapid Changes in Use by Herbivorous Fish Communities. New Horizons in Conservation Conference, March 29- 31, 2022. Virtual Oral Presentation.

Flores, J.A., Hayes, H., Williams, L., Fong, C.R., Holbrook, S.J., Schmitt, R.J., Barber, P.H., Fong, P. Physical Disturbance to Foundational Macroalgal Communities on Coral Reefs Causes Rapid Changes in Use by Herbivorous Fish Communities. Ocean Sciences Meeting, February 24-March 4, 2022. Virtual Oral Presentation.

Flores, J.A., Hayes, H., Williams, L., Fong, C.R., Holbrook, S.J., Schmitt, R.J., Barber, P.H., Fong, P. Physical Disturbance to Foundational Macroalgal Communities on Coral Reefs Causes Rapid Changes in Use by Herbivorous Fish Communities. Guest Lecture for UCLA Spring Marine Biology Quarter, April 4, 2022. Virtual Oral Presentation.

Flores, J.A., Hayes, H., Williams, L., Fong, C.R., Holbrook, S.J., Schmitt, R.J., Barber, P.H., Fong, P. Physical Disturbance to Foundational Macroalgal Communities on Coral Reefs Causes Rapid Changes in Use by Herbivorous Fish Communities. Guest Lecture for UCLA Winter Marine Biology Quarter, January 6, 2022. Virtual Oral Presentation.

Dellinger, R., **Flores, J.A.**, Rumberger, C., Armstrong, M., Benson, B.E., Bay, R. Assessing the Drivers of Genetic Diversity on Coral Reefs. UC Davis Undergraduate Research Conference, April 28-May 1, 2021. Virtual Oral Presentation.

Flores, J.A., Nelson, F., and Todgham, A.E. Thermal Tolerance and Heat Hardening Ability Between Sexes in *Gambusia affinis*. Western Society of Naturalists Conference, November 5-8, 2020. Virtual Poster Presentation.

Flores, J.A., Nelson, F., and Todgham, A.E. Thermal Tolerance and Heat Hardening Ability Between Sexes in *Gambusia affinis*. UC Davis Undergraduate Research Conference, April 28-May 1, 2020. Virtual Poster Presentation.

Flores, J.A., Nelson, F., and Todgham, A.E. Thermal Tolerance and Heat Hardening Ability Between Sexes in *Gambusia affinis*. UC Davis Educational Enrichment & Outreach Programs (EEOP) Annual Poster Symposium, Davis, CA, August 29, 2019. Poster Presentation.

Southeast Asia Climate Adaptation, Resilience, and Engagement (SEACARE) Program. Naga, Camarines Sur, Philippines, February 28-March 3, 2025. Professional Development.

Ecological Society of America Annual Meeting, Long Beach, CA, August 4-8, 2024. Professional Development.

National Academies of Science, Engineering, and Medicine Climate Crossroads Summit, Washington, D.C., July 16-17, 2024. Professional Development.

Western Society of Naturalist Conference, Monterey, CA, November 9-12, 2023. Professional Development.

UCLA Science Outreach Symposium, Los Angeles, CA, October 20. Professional Development.

Yale New Horizons in Conservation Conference, New Haven, CT, May 10-13, 2023. Professional Development.

AFFILIATIONS AND MEMBERSHIPS

United Nations Ocean Decade Early Career Ocean Professional (ECOP)	2025 – Present
American Genetic Association (AGA)	2025 – Present
Ecological Society of America (ESA)	2024 – Present
Center for Diverse Leadership in Science (CDLS)	2021 – Present
Association for the Sci. of Limn. and Ocea. (ASLO)	2021 – Present
Soc. for Adv. of Chic./Hisp. & Native Am. in Sci. (SACNAS)	2021 – Present
Western Society of Naturalists (WSN)	2020 – Present
Leaders of Achievement in Biological Sciences (LABS)	2019 – 2021
Bio. Undergraduate Scholars Program Honors Research (BUSP-HR)	2019 – 2021
Bio. Undergraduate Scholars Program (BUSP)	2017 – 2019

LEADERSHIP & SCIENCE OUTREACH

Career Panelist – UCLA Ocean Resources for Conservation and Advocacy

Oct. 2024

Shared my undergraduate and graduate journey as well as resources for undergraduate students interested in pursuing intramural research experience. Answered questions pertaining to my path towards a career in marine science research and how I am forging my path to ensure conducting a science that uplifts my community and those around me.

Participant and Co-Author – UC Queer & Trans Inclusive Field Summit

June 2024

UC faculty, postdocs, researchers, and graduate students from all campuses who do ecological field research, with the goal of building a supportive community and sharing resources. Presented and engaged with plenaries about the intersection of identity and ecological fieldwork and partake in community-building activities. Synthesized and co-authoring a manuscript to discussions to highlight the best practices and synthesize a list of resources that are most critical to LGBTQ+ field safety.

Graduate Student Panelist – UCLA Ecology and Evolutionary Biology Undergraduate Association

Feb. 2024

Shared my experiences and perspective as a student that directly pursued graduate school right after undergrad. Answered questions pertaining to my path towards a career in marine science research and how I am forging my path to ensure conducting a science that uplifts my community and those around me.

EEB Grad Peer Mentorship Board Member – UCLA Department of Ecology and Evolutionary Biology

June 2024 – Present

Facilitate graduate peer mentor and mentee relations within the EEB Department. Attended monthly meetings to update and discuss progress on upcoming events and action items.

EEB Grad Peer Mentor – UCLA EEB Grad Peer Mentorship Program

Nov. 2023 – Present

Scheduled interval meetings with a first year PhD student in the department. Provided mentorship, advice, and resources to answer any worries, questions, or concerns they had pertaining to student life in the department, institution, and the greater Los Angeles community.

K-12 Outreach Group – UCLA Center for Diverse Leadership in Science

Sept. 2021 – Present

Organize and participated in science enrichment activities with schools in the local Los Angeles area. Presented a webinar on the topic of food chain and food webs to students at El Dorado Elementary School.

Math & Sciences Academy – UC Early Academic Outreach Program

May 2022; May 2023

Organized volunteers to help facilitate the activities for the Marine Biology Module of the program. Designed presentation and hands on activities on shark physiology and technology used in marine sciences tailored to high school students in the LA County Area.

Pen Pal – Letters to a Pre-scientist

Sept. 2022 – Sept. 2024

Exchange letters with a middle school pen pal who shared similar science interests. I share my journey and experiences as a scientist and student in college and raise awareness of what a scientist and a career in science looks like.

Graduate Student Panelist – UCLA The Diversity Project

June 2022, June 2023, June 2024

Welcomed a new cohort of students to The Diversity Project. I shared my experience participating in the summer program, hurdles for LGBTQIA+ and BIPOC in navigating marine science, and graduate school as a queer trans person of color.

Student and Post Doc Experience Roundtable – California Ocean Science Trust

Jan. 2022

Participated in a series of roundtable discussions with students and faculty on obstacles and solutions to institutional change to cultivate a more inclusive, equitable and diverse academic culture. The roundtable resulted in a proceedings report titled “[Pathways to Advance DEI in California Coastal and Ocean Sciences](#)”.

Student Panelist – UC Davis Student Learning Outcomes Symposium

Sept. 2020

Invited to speak at the UC Davis Student Learning Outcomes Symposium to share my thoughts, experiences, and suggestions to UC DAVIS faculty on student learning in virtual classes and effects of COVID-19 on student learning.

Pen Pal – BUSP and Lee Middle School

Sept. 2018 – June 2021

Shared my experiences as a scientist and student in college through exchanging letters with students from Lee Middle School in Woodland, CA through a pen pal program through BUSP. Lead a campus tour for students from Lee Middle School.

CERTIFICATIONS

American Academy of Underwater Sciences (AAUS) Scientific Diver
National Association of Underwater Instructors (NAUI) Master SCUBA Diver
NAUI Advanced Rescue Diver
CPR and First Aid

SKILLS

Computer Skills

Proficiency with Microsoft Office (Word, Excel, and PowerPoint), ImageJ, QIIME2, and R. Website design through Wix.

Laboratory Skills

DNA Extraction
Polymerase Chain Reactions (PCR)
Gel Electrophoresis
DNA Gel Purification
Critical Thermal Maximum Methodology
Hypoxia Tolerance Test

Anxiety behavior test
Anti-predator response test
Fish husbandry
Fish dissections
Bacterial transformations
Bacterial inoculation via solid and liquid cultures

Field Skills

Identifying invertebrates of the major invertebrate phyla using comparative morphology, natural history, ecology, and behavior of living invertebrates.

Identifying Indo-Pacific reef fishes using morphology, natural history, ecology, and behavior.

Handling organisms from freshwater and marine environments.